FedCode: Communication-Efficient Federated Learning via Transferring Codebooks

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01 Context & approach

The challenge

Clustering model weights reduces value storage, but transmitting every weight index still costs bandwidth. *FedCode* usually exchanges only cluster centers in both directions. Periodic calibration also transfers compressed assignments to realign client and server models.

Our contribution

Reduce upstream and downstream traffic by exchanging compact codebooks.
Map existing weights to received centers without sending every index each round.
Use periodic calibration to control the communication-accuracy trade-off.

1 Broadcast centers

Cluster global weights and send sorted centers. Clients map existing weights to their nearest received centers.

2 Train and return codebooks

Clients train and cluster the full model. The server concatenates received codebooks and maps its weights to the nearest centers.

3 Calibrate periodically

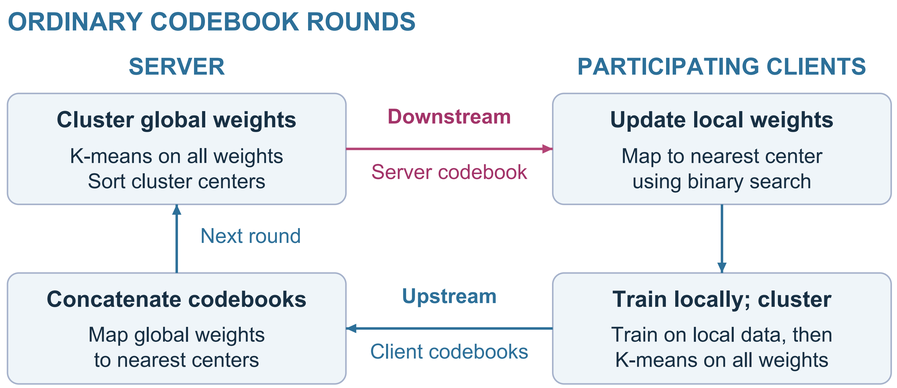
Also transfer compressed weight indices on calibration rounds. Reconstruct models; server-side calibration then applies FedAvg.

Scope

Communication volume is measured; client latency, energy and run-to-run uncertainty are not established. Calibration direction labels conflict in the paper; F1/F2 are retained without assigning directions.

02 Method

Exchange codebooks in both directions and periodically calibrate weight assignments.



Codebook = cluster centers; ordinary rounds omit weight indices.

PERIODIC CALIBRATION — SELECTED ROUNDS

Transfer codebook + weight indices in the relevant direction.

Clients reconstruct weights from indices and centers.

The server reconstructs client models, then applies FedAvg.

Core formulation

$$j^* = \arg\min_j |w_i - c_j|, \qquad w_i \leftarrow c_{j^*}$$

w_i: existing weight; c_j: received center. Ordinary rounds choose the nearest center; the server uses concatenated client codebooks.

$$\mathrm{DTR} = \frac{2Pb}{2Kb + (F_1 + F_2)P\lceil \log_2 K \rceil}$$

P weights, K centers, b bits/weight; F1 and F2 are calibration frequencies. Less index transfer reduces traffic.

Paper Code

03 Results

14.6×
joint communication reduction
IID CIFAR-100; ResNet-20.

58.07%
FedCode test accuracy
FedAvg: 61.22% on the same workload.

Compress both communication directions

Method	Accuracy (%)	Down DTR	Up DTR	Joint DTR
FedAvg	61.22	—	—	—
OMC-T	61.21	—	2.9	1.4
FedCompress	59.67	4.11	4.11	4.11
FedZip	57.98	—	19.8	1.9
FedCode (Ours)	**58.07**	**24.2**	**10.4**	**14.6**

IID CIFAR-100, ResNet-20; 10 clients, full participation, 100 rounds, 4 local epochs. *FedCode*: 64 centers, F1=0.2, F2=0.5. DTR = FedAvg bits / method bits; higher is better.

Calibration controls the trade-off

F1 / F2	Accuracy Δ (pp)	Joint DTR	Data (MB)
1 / 1	-0.86	5.3	394.21
0.2 / 0.5	-3.15	14.6	143.92
0.1 / 0.1	-21.64	44.3	47.35
0 / 0	-46.88	264.7	7.9

Same CIFAR-100/ResNet-20 setting; 64 centers. Accuracy change is relative to FedAvg; negative means lower accuracy. FedAvg communication: 2102 MB.

Takeaway

FedCode achieves **14.6× less joint communication** at 58.07% accuracy, 3.15 percentage points below FedAvg in this setting. Without calibration, communication falls further but accuracy drops 46.88 points below FedAvg.